One Sample testing Procedure

1) test μ known σ — test μ unknown σ, $df = n-1$ — test p

2) H_0: Old belief — H_a: New claim — complement of each other

always contain equality

test stat 3)

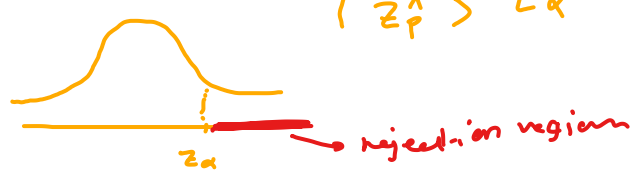
$$Z_{\bar{x}} = \frac{\bar{x} - \mu_{H_0}}{\sigma/\sqrt{n}} \qquad t_{\bar{x}} = \frac{\bar{x} - \mu_{H_0}}{s/\sqrt{n}} \qquad Z_{\hat{p}} = \frac{\hat{p} - p_{H_0}}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}}$$

Z_α : $1-\alpha$ Percentile (i.e. 0.05 → 95 percentile)

Rej Region 4)

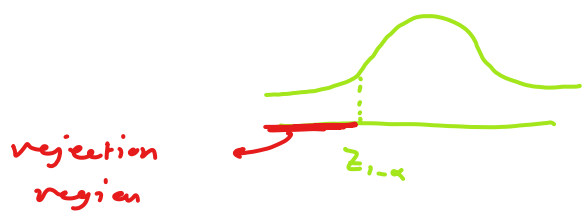
Right tailed test (H_1: $\mu > \mu_0$, $p > p_0$) Reject if

$$\begin{cases} Z_{\bar{x}} > Z_\alpha \\ t_{\bar{x}} > t_{\alpha, n-1} \\ Z_{\hat{p}} > Z_\alpha \end{cases}$$



Left tail test : (H_1: $\mu < \mu_0$, $p < p_0$) reject if

$$\begin{cases} Z_{\bar{x}} < Z_{1-\alpha} \\ t_{\bar{x}} < t_{1-\alpha, n-1} \\ Z_{\hat{p}} < Z_{1-\alpha} \end{cases}$$



test two-tailed : (H_1: $\mu \neq \mu_0$, $p \neq p_0$) Reject if

$$\begin{cases} |Z_{\bar{x}}| > Z_{\alpha/2} \\ |t_{\bar{x}}| > t_{\frac{\alpha}{2}, n-1} \\ |Z_{\hat{p}}| > Z_{\alpha/2} \end{cases}$$